

# Repeatable tool changes for ioSender + grblHAL

*A turnkey tool-change and work-setup package for CNC newcomers.*

This package turns the fiddliest, most error-prone parts of running a job — setting work zero and getting tool lengths right — into two things you can do without thinking:

1. **Start a job:** install the 3D probe, press OK a couple of times. The machine finds the corner of your stock and sets work zero by itself.
2. **Change a tool mid-job:** when the program calls for a new tool, the machine parks, says “Swap to T3, press cycle start”, you change the bit and press cycle start. The machine measures the new tool and carries on.

That’s it. No paper shims, no “jog down until it just touches”, no re-zeroing Z every time you change a bit.

## Why you want this

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The single biggest reason is **repeatability**: *your second run matches your first*.

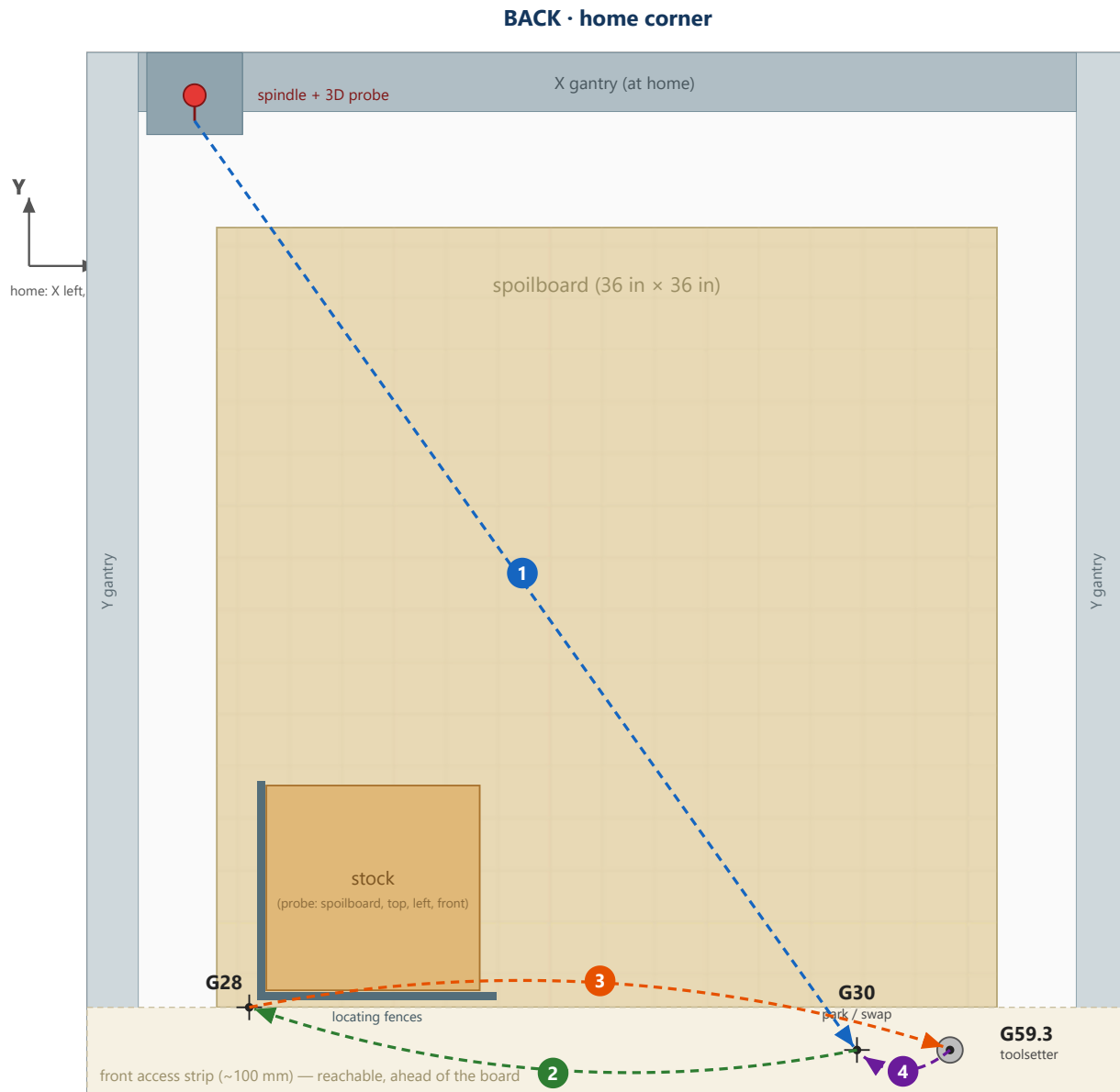
On a bare machine, every tool change means you manually re-establish how far the new tool sticks out — usually by jogging down onto the work and eyeballing it. Do that ten times in a job and small errors creep in: a step in a surface, a pocket that’s 0.2 mm too deep, a scrapped part. Do the same job next week and the numbers are different again.

This package removes the human from that loop:

- A **toolsetter** (a touch puck bolted to your table) measures every tool against the *same fixed point*, every time. Because the machine is **homed**, that point is in the same place after a power cycle, tomorrow, next month.
- The **first tool** of a session (your 3D probe) sets the reference. Every tool after it is automatically offset relative to that reference, so Z is consistent across all tools and all sessions — with **no manual touch-off, ever**.
- A **3D probe** finds the corner of your stock and sets X, Y and Z work zero by touching it, instead of you nudging the bit up to an edge by eye.

The payoff: setup is faster, mistakes are rarer, and you can re-run a job and trust the result. Your job action shrinks to **change the bit and press a button**.

## The big idea, in one picture



### Probe-and-park sequence (from home):

- ➔ 1 · Home → G30 — rapid to park; operator installs the 3D probe.
- ➔ 2 · G30 → G28 — probe spoilboard, stock top, left & front faces → set G54 work zero.
- ➔ 3 · G28 → G59.3 — probe the toolsetter puck with the 3D probe → set the TLO reference.
- ➔ 4 · G59.3 → G30 — return to park and wait; operator starts the milling job.

Top-down view (48" × 48" machine). Z parks high throughout; rapids shown as dotted legs.

Two probes, two jobs:

- The **3D probe** answers *"where is my stock?"* (work zero).

- The **toolsetter** answers “*how long is this tool?*” (tool length), and keeps every tool consistent with the reference.

Homing ties it together: it’s what makes “the same fixed point” actually be the same point every time you turn the machine on.

## What you need

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### Hardware

- A **3D touch probe** (edge/corner finder) wired to the controller’s main probe input.
- A **toolsetter / tool-length puck** (a normally-closed touch switch) wired to the controller’s toolsetter input, bolted somewhere permanent on the table.
- A machine with **working homing** (limit switches / sensors).

### Firmware (grblHAL)

- Built with **expression support** ( `EXPR` ), a **filesystem** (LittleFS and/or SD card), and **macro tool-change** support (so `M6` runs `tc.macro` ).
- A **tool table configured** ( `N_TOOLS > 0` ) — the reserved probe tool number is derived from it.
- `$22` (homing-on-startup) enabled is strongly recommended — it’s the primary safety net that stops the machine moving before it’s homed.

ioSender ships the two controller-side macros this package needs ( `tc.macro` , `probe_tfl.macro` ) and can install them for you (see setup).

## One-time setup

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You do this once. After that, day-to-day use is just the everyday workflow below.

### 1. Mount the toolsetter

Bolt the toolsetter puck somewhere it will live permanently and where any tool can reach it without hitting a clamp or fixture. Its location is fixed forever after — that permanence is what makes tool lengths repeatable.

### 2. Teach the machine three positions

These are stored machine positions the macros rapid to. Set each by jogging the spindle to the spot, then storing it. You only do this once (re-do only if you move the toolsetter or your work-holding corner).

| Position     | What it is                                                                                                                                                                    | How to set it                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>G59.3</b> | Directly over the toolsetter puck, at a safe height where the longest tool clears the puck on the way in.                                                                     | Jog over the puck at safe Z, then in the MDI: <b>G10 L20 P9 X0 Y0 Z0</b> (or use ioSender's Offsets view to set coordinate system <b>G59.3</b> ). |
| <b>G30</b>   | The tool-swap / park position — somewhere comfortable to reach the spindle, near the top of Z. Usually near the toolsetter.                                                   | Jog there, then in the MDI: <b>G30.1</b> .                                                                                                        |
| <b>G28</b>   | The corner-probe approach: just outside the front-left corner of where your stock sits, 10–30 mm clear in both X and Y, with open air below the probe down to the spoilboard. | Jog there, then in the MDI: <b>G28.1</b> .                                                                                                        |

**Tip:** If you always place stock against the same corner stop or fixture, **G28** can stay set forever too. If your stock position varies, re-set **G28** before Start Job so it lands just outside the new corner.

### 3. Install the controller macros

In ioSender: **File** → **Controller macros....** The dialog lists the bundled macros and whether each is already on the controller. Click **Upload all missing** to copy them into the controller's persistent LittleFS store.

These are **example** macros — they assume the positions and probe geometry described here. Read "Adapting to your machine" below before your first real run, especially if your probe or wiring differs.

If you tick **"Offer to install missing macros when I connect"**, ioSender will prompt you automatically on future connections whenever something's missing.

### 4. Reserve the top tool number in your CAM

The **highest tool number** — **T** = your configured tool count (**N\_TOOLS**) — is reserved as the **reference-probe sentinel**: it tells the tool-change macro "the 3D probe is in the spindle." The macros read this number at runtime (**#<\_tool\_table\_size>**), so it adapts to your machine automatically — on a controller built with **N\_TOOLS = 8** the reserved number is **T8**, with **N\_TOOLS = 32** it's **T32**. **Never assign that top number to a real cutting tool** in your CAM/post; use lower numbers for cutters.

## 5. Create the Start Job macro

Create an ioSender macro called **Start Job** whose body is a reference to the bundled `start_job.macro` file (a single `@<path-to>/start_job.macro` line). This lets you edit the file and re-run without copy/paste. (See ioSender's macro manager.)

## Everyday workflow

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### Starting a job

1. Home the machine (or have it homed on startup).
2. Run the **Start Job** macro. It will:
  - Check prerequisites (homed, idle, `G28` / `G30` / `G59.3` set, expressions available) and stop safely if anything's missing.
  - Ask for your **3D probe tip radius**.
  - Park at `G30` and ask you to **install the 3D probe**.
  - Rapid to `G28` and probe the **front-left corner** of your stock — setting X, Y and Z work zero (`G54`) automatically.
  - Park at `G30` and ask you to **leave the probe in the spindle**.
  - Touch the probe on the toolsetter to record the **reference** the rest of the job measures against.
3. You now have work zero set and the tool-length reference established, with the probe still in the spindle, parked at `G30`.

### Running the program

Load your cutting program and run it. The **first** `M6` (tool change) does the real work for you:

1. Spindle and coolant stop; Z lifts to a safe height.
2. The machine parks at `G30` and shows **"Swap to T3, press cycle start"**.
3. You **remove the probe, install tool T3**, and press **cycle start**.
4. The machine moves over the toolsetter (`G59.3`), measures the new tool, and applies the right length offset automatically.
5. It returns to `G30` and the program continues cutting.

Every subsequent tool change is the same three steps: **swap the bit, press cycle start, walk away**. Z stays correct across every tool because each one is measured against the same reference.

## What the safety gates do

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The Start Job macro refuses to run — with a clear message — unless:

- the machine is **homed** (so all those stored positions mean what they should),
- the controller is **idle**,
- **G28**, **G30** and **G59.3** are **set**, and
- the firmware supports **expressions**.

This is deliberate: the macros rapid to stored positions, so if those positions aren't set, the macro stops *before* moving rather than rapiding somewhere unexpected. Homing-on-startup ( **\$22** ) backs this up at the firmware level.

## How it works under the hood (optional reading)

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You don't need this to use the package, but here's what's actually happening.

**The reference tool.** The 3D probe runs as the **top tool number** ( **T** = N\_TOOLS). When **tc.macro** sees the selected tool equal to **#<\_tool\_table\_size>**, it knows the probe is in the spindle: it skips the swap prompt, probes the puck through the *main probe input* (the sensitive 3D-probe stylus, not the puck's switch), and stores the probe's measured puck height as the session reference ( **#<\_tlo\_ref>** ). No offset is applied for the reference itself.

**Each later tool.** For any other tool number, **tc.macro** parks at **G30**, prompts the swap, and then moves to **G59.3** to probe the puck through the *toolsetter input* (the puck's normally-closed switch fires when the rigid tool pushes on it). It then applies a dynamic tool-length offset:

```
G43.1 Z[ this_tool_puck_Z - reference_puck_Z ]
```

So every tool is expressed as a difference from the reference tool. Because the puck is fixed and the machine is homed, those differences are consistent run to run — that's the repeatability.

**Work zero.** **probe\_tfl.macro** probes the spoilboard, the stock top, and the left and front faces near the front-left corner, then sets **G54** X/Y/Z. It adapts its probe heights to the measured stock thickness, and corrects X/Y by the probe tip radius you entered.

**The three positions.** **G59.3** = over the puck (toolsetter), **G30** = tool-swap / park, **G28** = corner approach. **G53 Z-5** (just below the top of travel) is used as the safe transit height.

## Troubleshooting

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- **“Start Job aborts immediately.”** Read the message — usually a position ( `G28` / `G30` / `G59.3` ) isn’t set, or the machine isn’t homed. Re-do the one-time setup step for whatever it names.
- **The corner probe misses / crashes into stock.** `G28` is probably too close to (or inside) the stock corner. It must sit 10–30 mm *outside* the corner in both X and Y with clear air below. Re-set `G28`.
- **Tool change probes the wrong spot.** Re-check `G59.3` — it must be centered over the puck at a height the longest tool clears on the way in.
- **Tool lengths look off by a constant.** Make sure Start Job ran first (it sets the reference). Running a program’s `M6` without a reference set means there’s nothing to offset against.
- **Macros aren’t found by the controller.** Confirm they’re present: **File** → **Controller macros...** should show them as installed (LittleFS). Some controllers only pick up a newly-installed `tc.macro` after a reboot.

## Adapting to your machine

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The bundled macros are a known-good *starting point*, not a universal config. Before trusting them on your machine, check the assumptions:

- **Probe geometry.** `probe_tf1.macro` assumes a probe with an 8 mm-tall, 2 mm-diameter tip that flares to 4 mm above. A different stylus may need the probe heights adjusted.
- **Probe wiring / inputs.** It assumes the 3D probe is on the main probe input and the toolsetter on the toolsetter input, with correct polarity ( `$6` ).
- **Work-holding corner.** The corner routine probes the **front-left** corner. If you reference a different corner, the probe directions need changing.
- **Stock thickness.** Reliable side-probing needs stock at least ~3 mm thick.
- **Tool table.** The reserved probe tool is the top slot, so the controller needs a tool table ( `N_TOOLS > 0` ).

To change any of these you edit the macro files and re-upload them via **File** → **Controller macros...** (remove the old copy, then upload the edited one). The on-machine copies live in LittleFS so they survive power cycles and firmware updates.

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*Reserve the top tool number (= `N_TOOLS`) for the probe. Home before you start. Then: change the bit, press the button.*